

THERMABRICK · DESIGN DECISION RECORD

ThermaBrick recommendations accepted

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DOCUMENT CONTROL

Document ID	TBK-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/thermabrick		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation.

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record covers every ThermaBrick item that was "Proposed, awaiting Amish" and carried a recommendation, in `docs/REVIEW.md` (the September 2026 review and the 2026-09-25 rein-in session) and in TBK-PRC-001, Table 4. ThermaBrick has no TBK-DDR-001; the design proposals D1 to D8 were never written up as a separate record, so this record also serves as their design decision record. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation stay open.

The portfolio stays at TRL 3. Any decision that needs building, buying, testing or measurement is recorded as decided but on hold, because TRL 4 is on hold by Amish's instruction. The archived test article material under `archive/out-of-phase-trl4/` is frozen and was not edited.

■ Decisions

Each item below is decided by Amish, 2026-09-25: go with recommendation.

No.	Item	Recommendation adopted	Effect in this repo
A1	Test article budget	Raise the accepted budget from \$635.70 to \$727.70, the BOM total after the build procedure's additions	<code>project.yaml</code> <code>budget_usd</code> 636 to 728; README budget line; TBK-PRB-001 v0.8, section 3 and section 5; TBK-REQ-001 v0.9, R20 restated; <code>bom/bom-notes.md</code> . On hold with TRL 4: nothing is bought
A2	Full-scale budget	Revisit the full-scale budget with the test article's results	Recorded in TBK-PRB-001, section 5 and TBK-PRC-001, section 9. On hold with TRL 4
A3	D1 to D8, full-scale design (TBK-PRC-001, Table 4)	Accept all eight: silica sand, 150 to 450 °C (D1); cartridge heaters in capped wells (D2); twelve 250 W wells (D3); closed U-tubes (D4); cold-side damper, mixing tee and heat trap (D5); wells end at the lid (D6); AES hot face with stone wool bulk (D7); 240 V, 3.0 kW, burst-fire SSRs and contactor (D8)	TBK-PRC-001 v0.5, Table 4 status and section 8 text. No change to the geometry, BOM or calculations, because the design already embodies D1 to D8
A4	Full-scale design choices never reviewed (REVIEW.md, section 2.3)	Accept the storage window with the 550 °C well-wall limit; twelve 250 W heater wells, 1-1/4 in U-tubes, 50 mm AES blanket plus 267 mm stone wool	Already in TBK-PRC-001, TBK-CAL-001, <code>cad/src/model.py</code> and <code>bom/bom.csv</code> . No change
A5	Requirement targets R1 to R20 (REVIEW.md, section 2.3)	Accept the targets as written, including R8's 500 °C drum-wall limit	TBK-REQ-001 v0.9 summary. R20 restated to follow A1
A6	Charge-time miss on R3	Choose among cutting the loss, twelve 1 in wells (8.4 h) or relaxing R3 to 10 h only after the test article has measured sand conductivity	R3 stays not met at 9.6 h against 8 h. TBK-REQ-001 R3 and TBK-PRC-001 sections 8 and 9 record the path. On hold with TRL 4
A7	Test article design choices (REVIEW.md, section 2.3): 16 US gal drum, 68 kg of sand, four heater cells, one U-tube, 120 V, stone wool hot face, 12 V blower with dilution stack	Accept as proposed	Archived; decided but on hold with TRL 4
A8	Controller enclosure: 250 × 200 × 150 mm polycarbonate, HDR-15-12 DIN-rail supply, DPDT relay, REX-C100 independent limit, ungrounded thermocouples	Accept as proposed	Archived; decided but on hold with TRL 4
A9	Firmware safety rules: heaters off until the logging host connects; any fault switches the heater mode off; T1 trips at 575 °C and the sand at 560 °C	Accept as proposed	Archived; decided but on hold with TRL 4
A10		Accept as proposed	

No.	Item	Recommendation adopted	Effect in this repo
	Test plan TBK-TST-001 pass criteria, tolerances and fitted power correction		Archived; decided but on hold with TRL 4
A11	Optional controller board (R-78E regulator, IRLZ44N, 2N3904)	Keep it optional and deferred, with the circuit choices as proposed	Archived; decided but on hold with TRL 4
A12	KiCad tooling	Install KiCad properly (<code>brew install --cask kicad</code>) before any schematic or board is regenerated	Needed only for the archived electronics; on hold with TRL 4

Table 1. Items decided by Amish on 2026-09-25.

Numbers before and after:

- `budget_usd` : 636 to 728.
- Test article budget: \$635.70 accepted to \$727.70 decided.
- TBK-REQ-001 status: 15 met, 3 open, 2 not met (R3, R20) to 16 met, 3 open, 1 not met (R3). R20 changed because its target was restated to the decided test article budget, not because the cost fell. The full-scale estimate stays \$3,674.

■ Still open

These items carried no recommendation and stay "Proposed, awaiting Amish".

- **Standby loss.** Whether 459 W at full charge (8.7 kWh, 47 % of the window, over 24 h idle) is acceptable, or whether to adopt the stainless leg sections and microporous panel (TBK-CAL-001, Table 5: 357 W for about \$450). TBK-PRB-001, section 5, item 2.
- **GFCI choice.** A 5 mA GFCI or 30 mA equipment ground-fault protection, to be chosen with the installing electrician (TBK-PRC-001, sections 8 and 9).

Confirmations that need data, not a decision, also remain: the fan's pressure curve, the stone wool base board's compressive strength, and every price in the BOM.

■ Consequences

- No geometry, BOM line or calculation changed, so `cad/src/model.py` , TBK-DWG-001 (Rev P1) and TBK-CAL-001 (v0.2) keep their revisions. They were regenerated only to refresh the title block footer.
- The repo stays at `trl: 3` and `trl_target: 3` .
- No cross-repo action arises from these decisions.